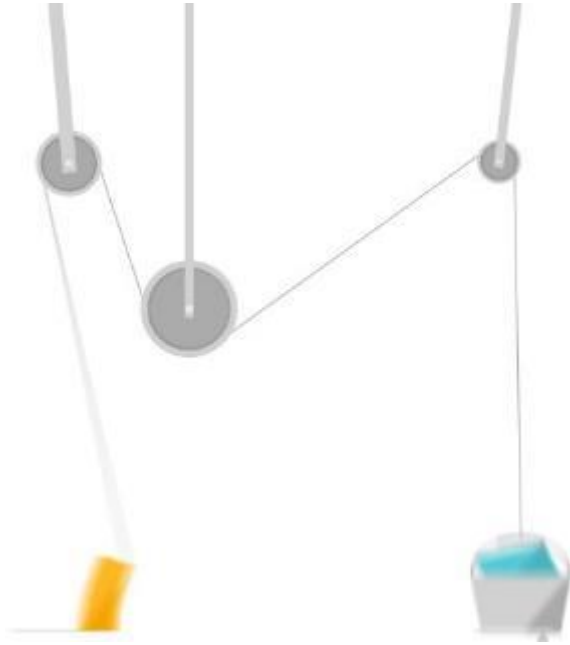
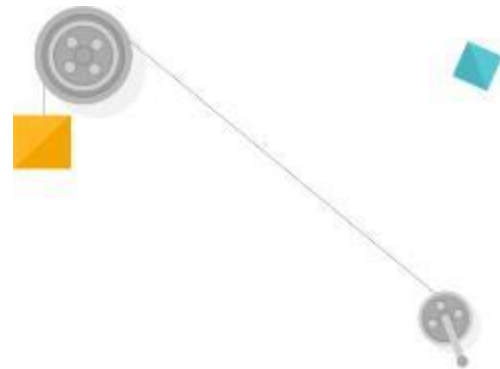




Simple Machines



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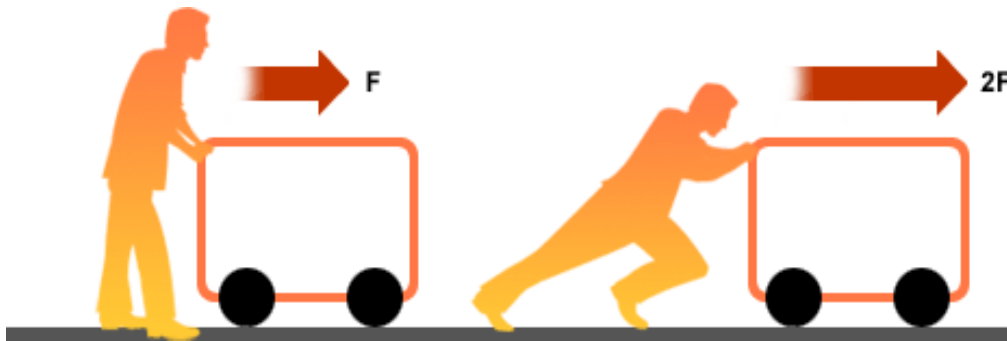
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SIMPLE MACHINES

A force is an action that causes objects to move. It can stop a moving object or change the direction of a moving object. A push or a pull is a force. Whenever, force is applied, energy is used.



If you want to shift a small object from one place to another it is possible, but if a large object needs to be moved it is difficult and requires a lot of effort.

When is work said to be done?

Work is done only when an object moves on applying force. When you run or walk you move, you do work. If you stand in the same place, you do not do any work.

We use machines to make our work easier. A device used to make work easier is called a machine.

For example, a washing machine makes the task of washing clothes easier.

A car makes the task of travelling to a faraway place easier.



Cycle



Tractor



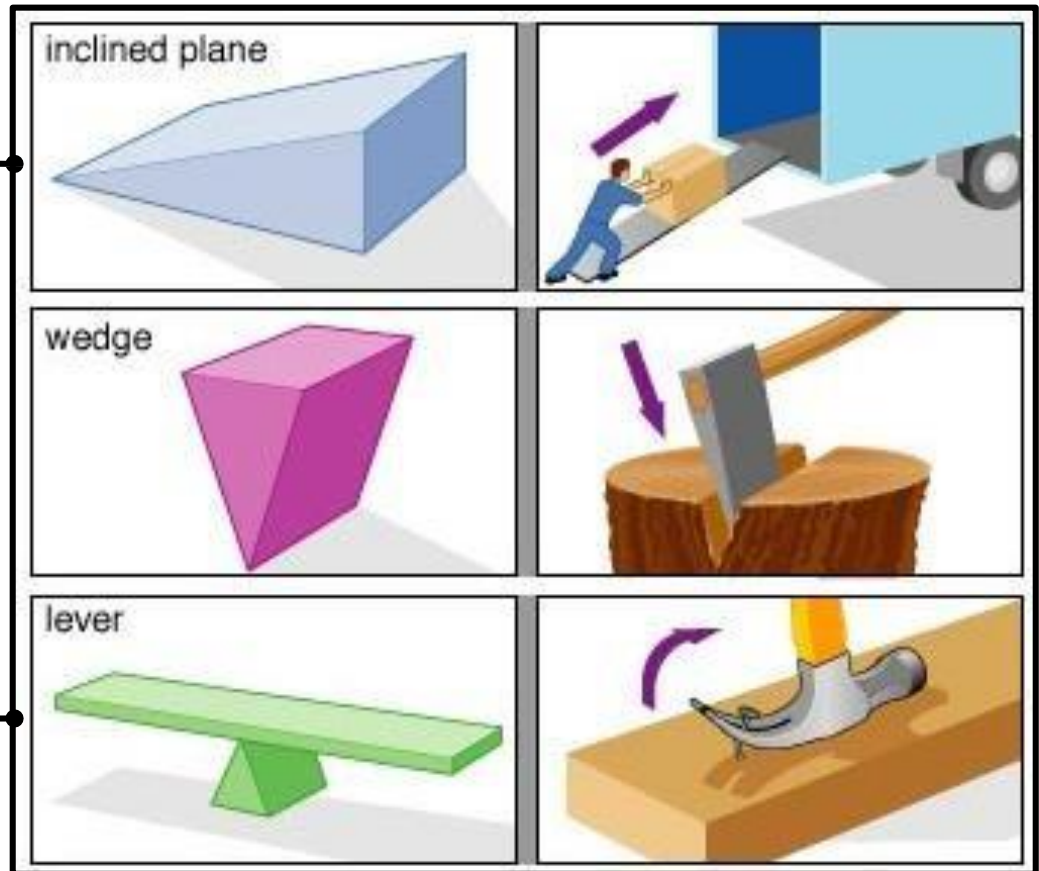
Mixer

TYPES OF MACHINES

Simple Machines:

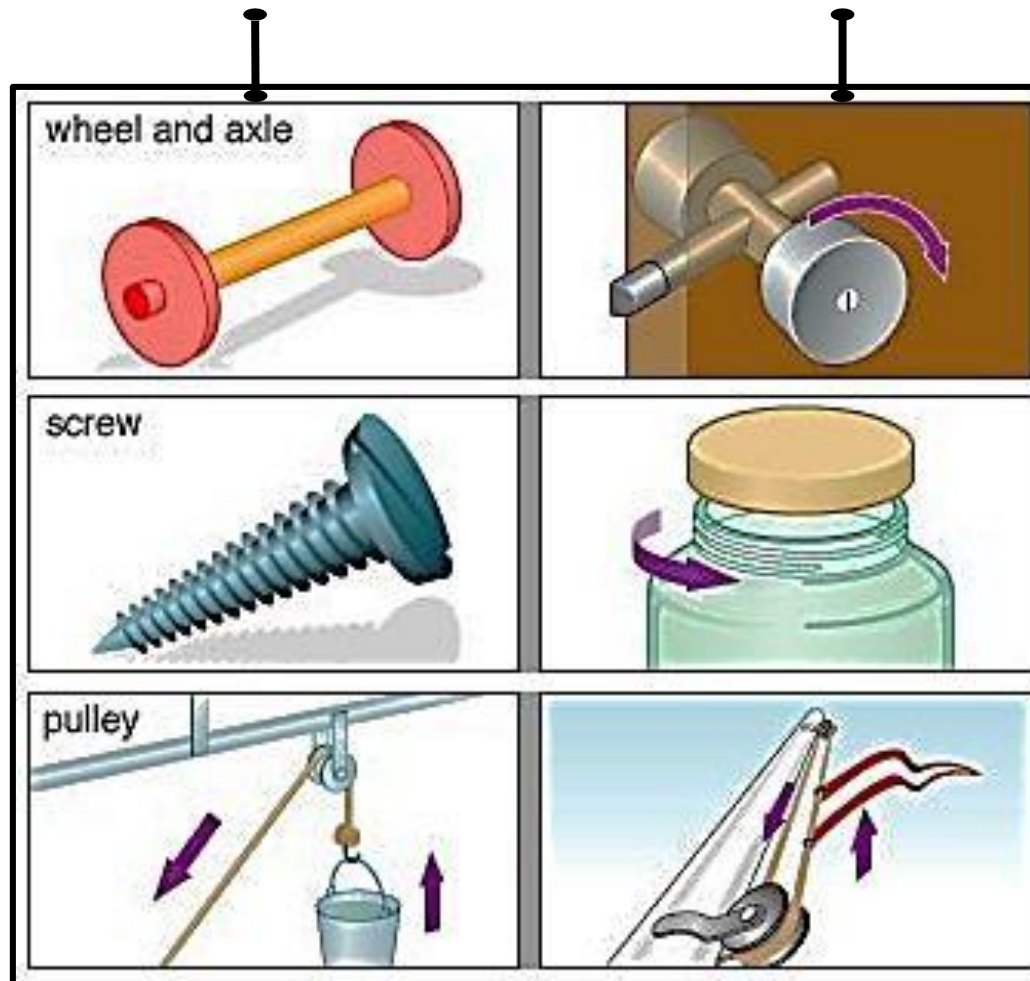
A machine need not be something complex like a car or a washing machine. Any tool that makes work easy is a machine.

A hammer makes a task of fixing nails easy. An axe makes the work of cutting wood easy. A needle makes the job of stitching clothes easy. All these tools are simple machines.



TYPES OF MACHINES

An inclined plane, wheel and axle, screw, wedge, lever and pulley are six types of simple machines.

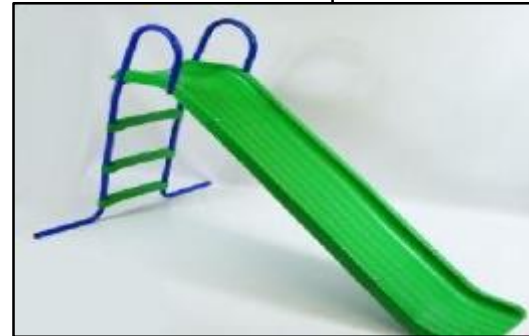


INCLINED PLANE

The effort required to push a heavy rock over a gradual slope is lesser than on a steep slope. A **gradual slope** is used to make work easier.

A gradual sloping surface is called an **inclined plane**.

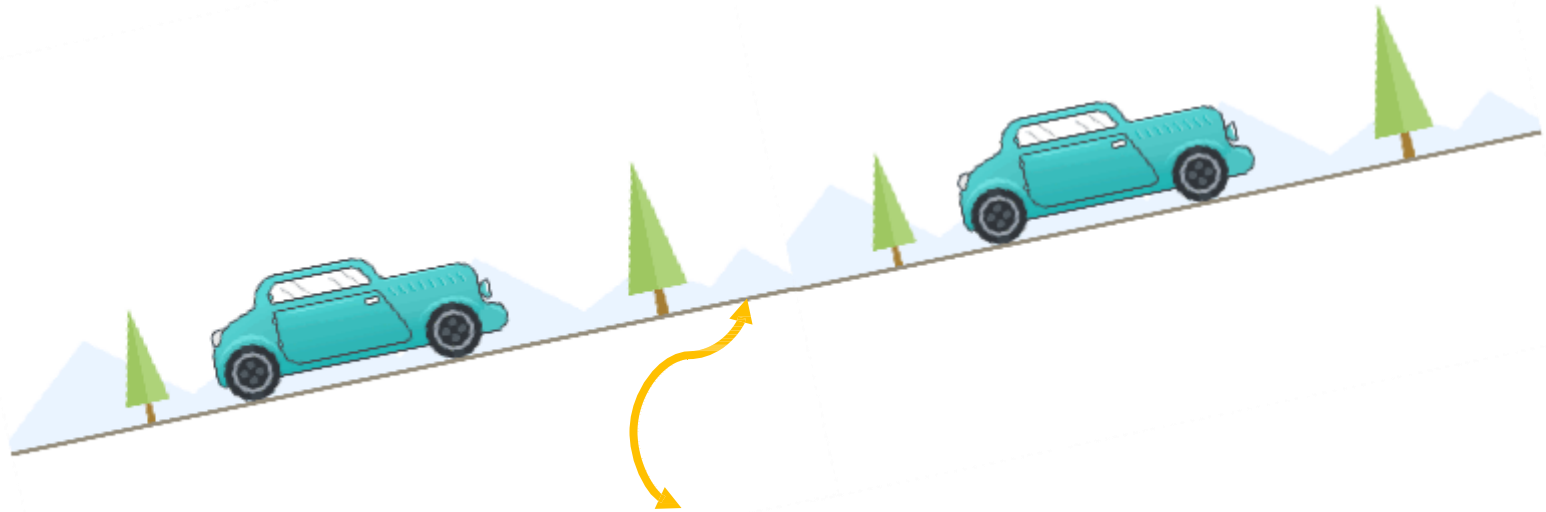
It is easier to move an object over an inclined plane. It is a simple machine that requires no moving parts because something or someone else exerts a force on the object being moved.



Since ancient times, inclined planes have been used to shift heavy objects to a higher level. A ramp, a flight of stairs, slides, escalators and ladders are examples of an inclined plane.

INCLINED PLANE

A winding road on a hill or a mountain is a gradual slope. It is not a steep slope. It is an inclined plane. The winding road, makes travelling through mountains or hills easier.



A gradual sloping surface is called an **inclined plane**.

SCREW - AN INCLINED PLANE

Do you know that a screw is a type of an inclined plane? A screw is a type of inclined plane wrapped around a rod. It resembles a winding road. The grooves on a screw are like the road on a hill.

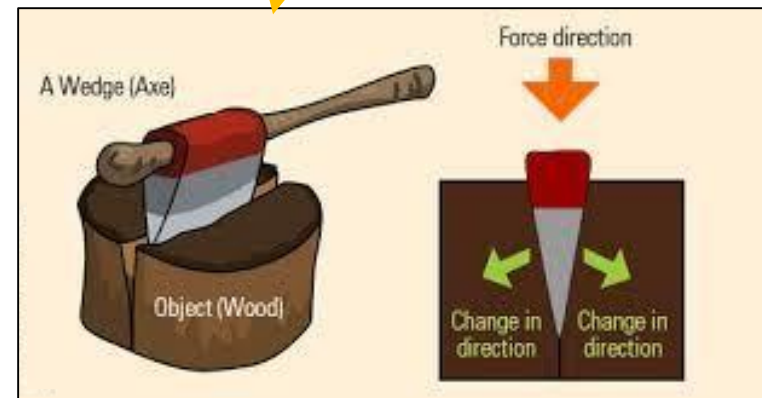


The grooves gradually rise from one end to the other. Every turn of a groove is a tiny inclined plane. It helps to move the screw through the wood with less effort. The grooves also help in providing a better grip. Hence, it is not easy to pull a screw out of a machine or a piece of wood.

WEDGE

A wedge is a triangular shaped tool which is an inclined plane. It is one of the six simple machines. It can be used to separate two objects or portions of an object, lift an object, or hold an object in place. A wedge looks like two inclined planes stuck together. The edge of a wedge is often called the blade.

One difference between a wedge and an inclined plane is that, while the inclined plane stays still, a wedge moves to do its work. To chop down a tree, someone must swing the axe. To cut out a paper heart, you need to move the scissor blades up and down.



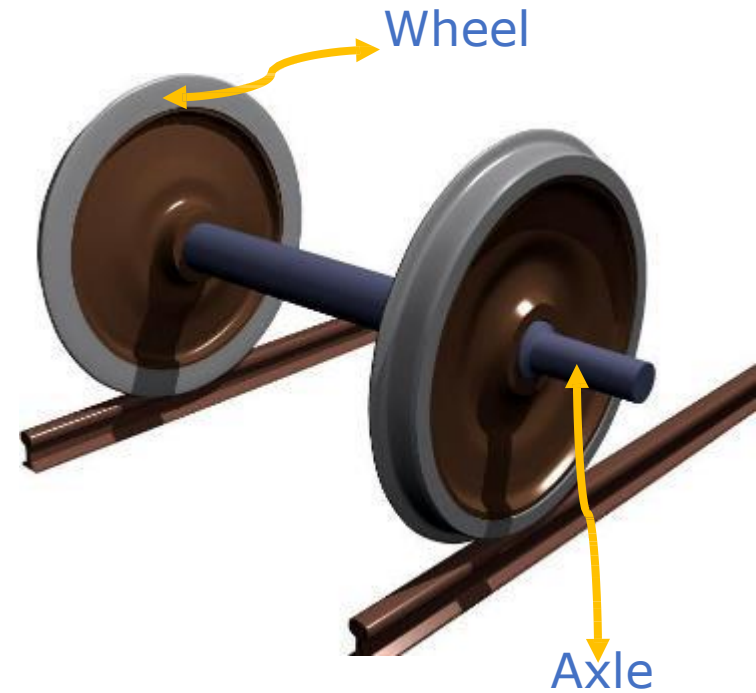
WHEEL AND AXLE

A wheel and an axle together form a single machine. A piece of rod attached to the centre of a wheel is called an axle.

The wheel and axle move together. One moves the other. Without an axle, a wheel cannot be used as a machine.

Wheels fixed to an axle move with less effort. They are used to move heavy objects and to set things in motion.

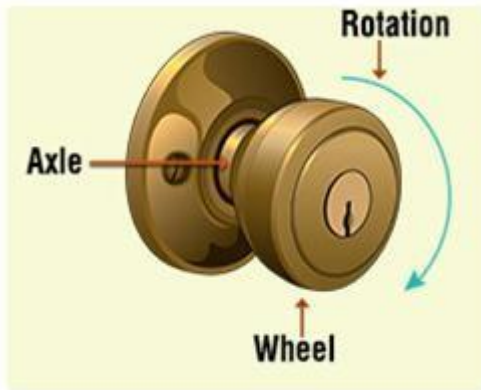
All vehicles move with the help of wheels and axles. Wheels and axles are part of a fan, a mixer grinder, a washing machine and a steering wheel of a car or a bus.



WHEEL AND AXLE

A screwdriver and the knob of a tap are examples of a wheel and an axle. Most machines have a wheel and an axle as one of their parts.

Doorknob



Fishing Rod



Steering Wheel



Cycle



Spanner

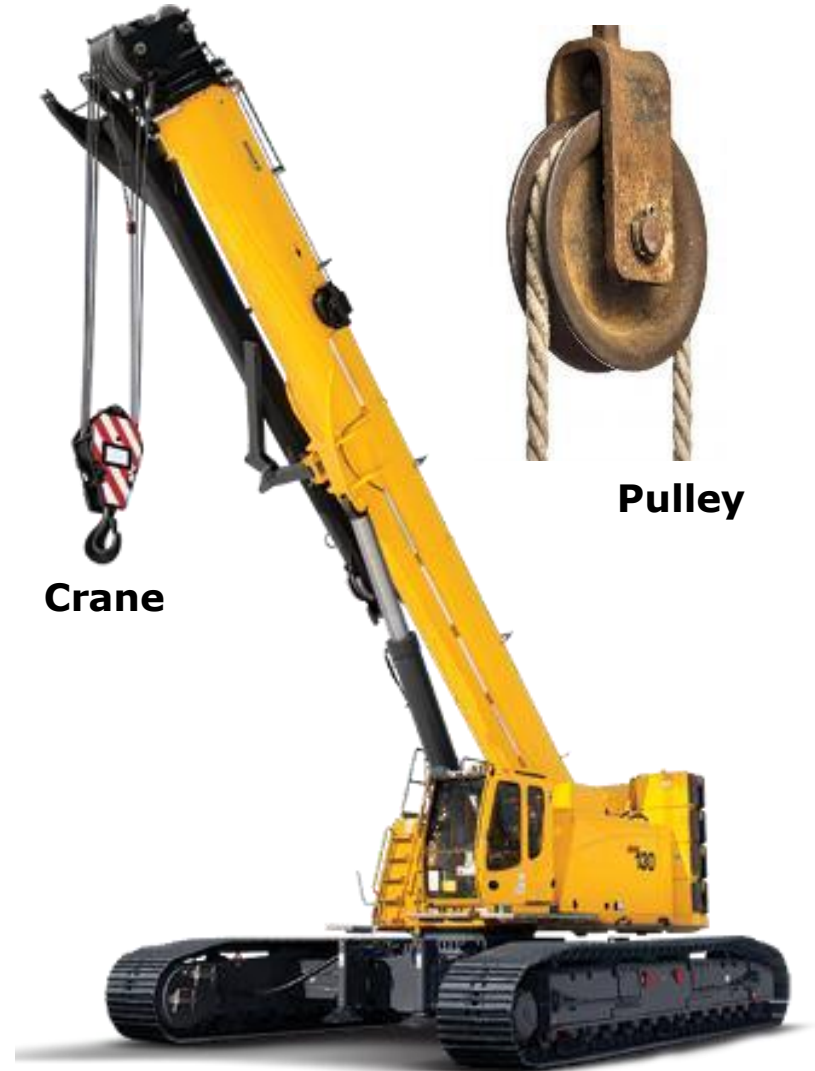


PULLEY

A pulley is a grooved wheel with a rope or cable around it. It can be movable or fixed to a support. This simple machine changes the direction of the force used. The most common use of a pulley is to fetch water from a well. The pot of water is pulled upward by pulling the rope downward.

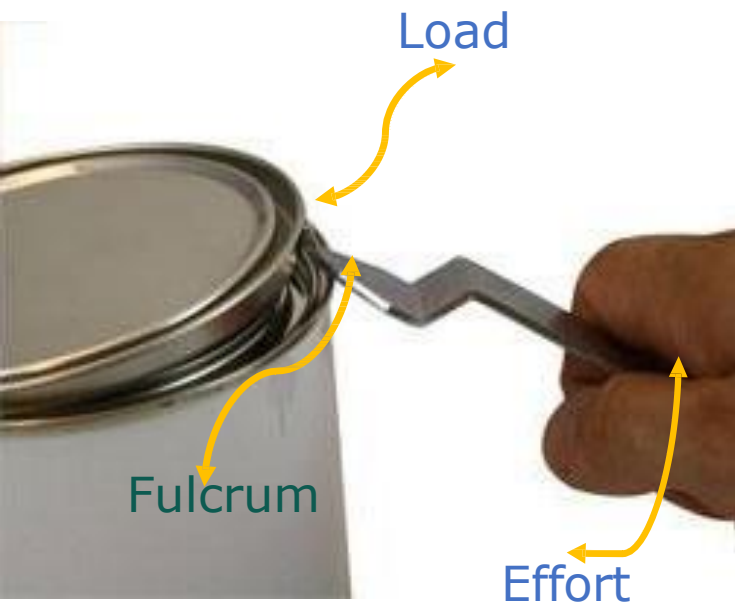
When more than one pulley is linked with the same cable around them, they help in lifting or lowering loads with very less effort.

A crane works on many pulleys and can shift extremely heavy objects.



LEVERS

Have you ever used a spoon to open a lid of a container? Did you know, you were using a spoon as a machine called a lever?

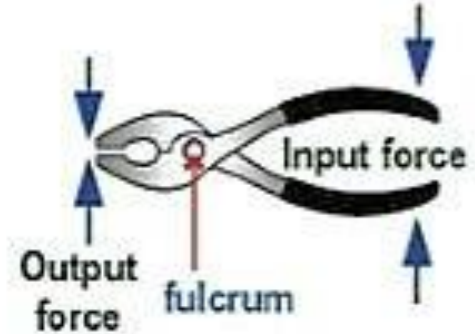


A lever is a bar that can be moved about a fixed point called a fulcrum. It is a simple machine because it helps in performing a task easily.

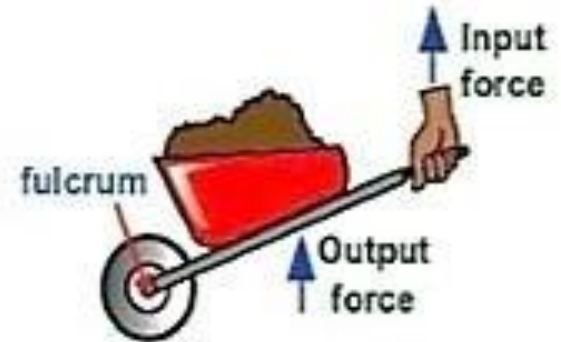
THREE TYPES OF LEVERS

The first-class lever, the second-class lever and the third-class lever.

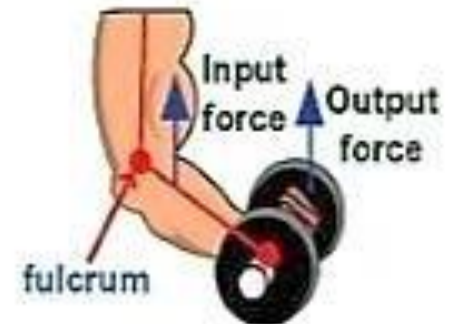
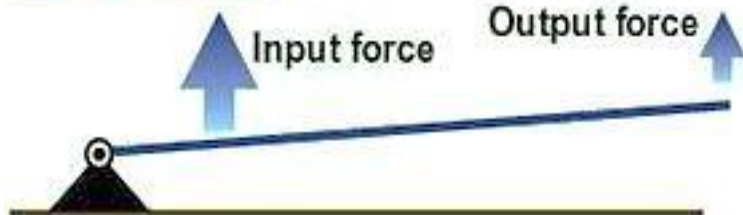
1st Class



2nd Class

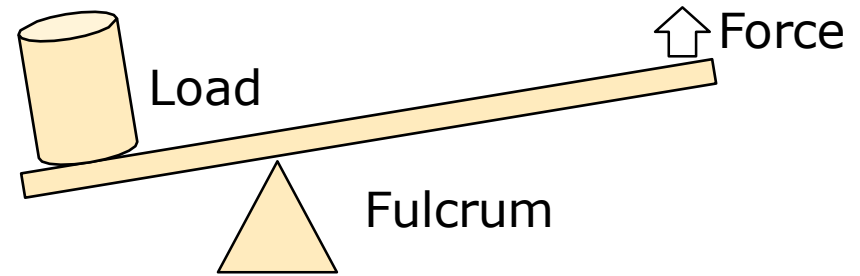


3rd Class

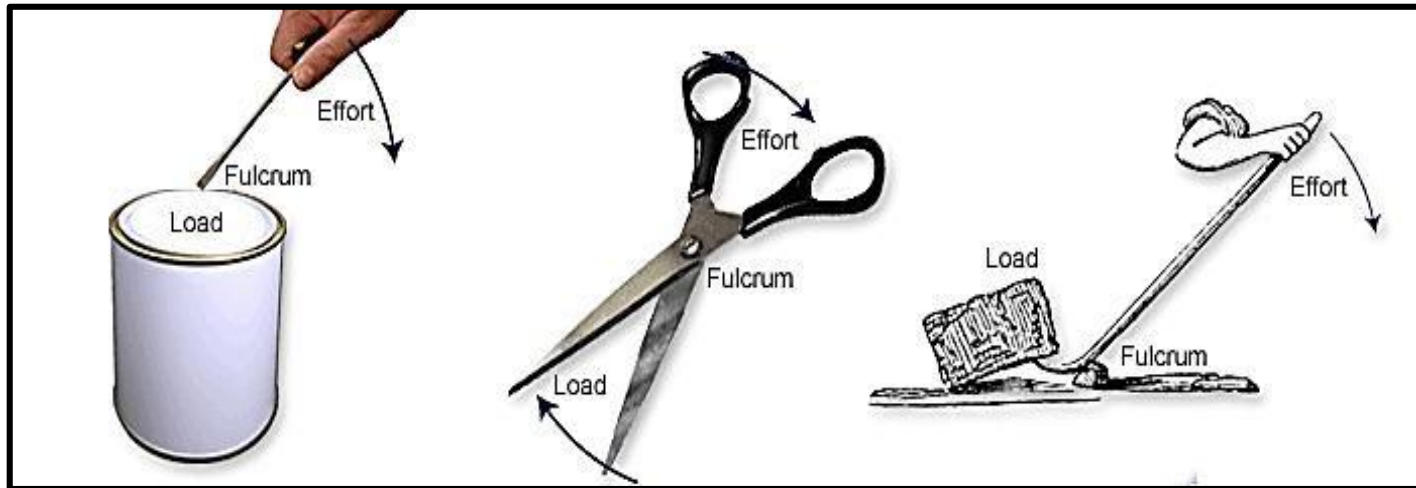


FIRST-CLASS LEVER

First-class lever

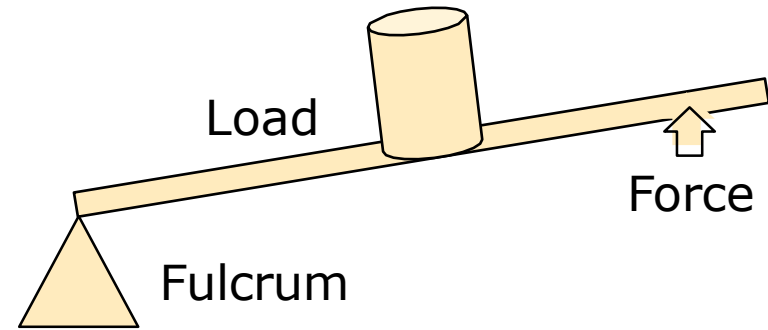


First-class lever: In the first-class lever, the fulcrum is in the middle. In this example, the spoon is the lever, and the fulcrum is the rim of the tin. The load is the lid of the tin and effort is made at the other end of the lever (spoon).

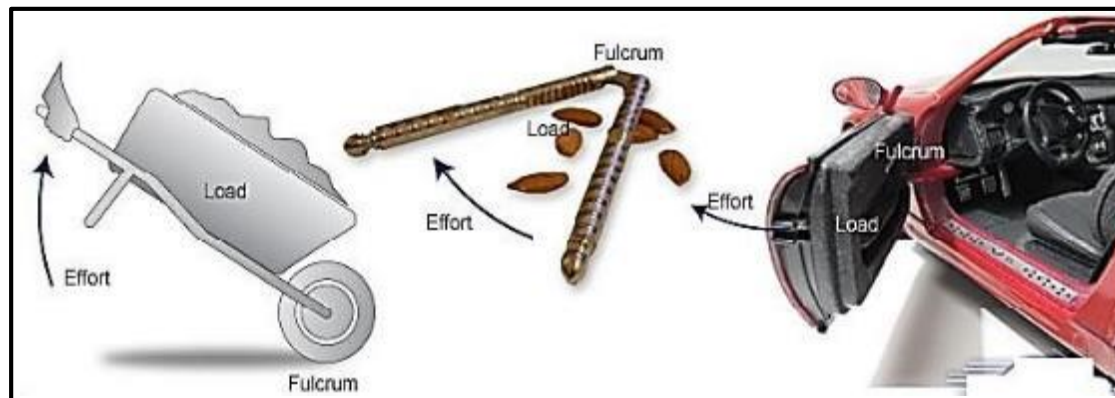


SECOND-CLASS LEVER

Second-class lever



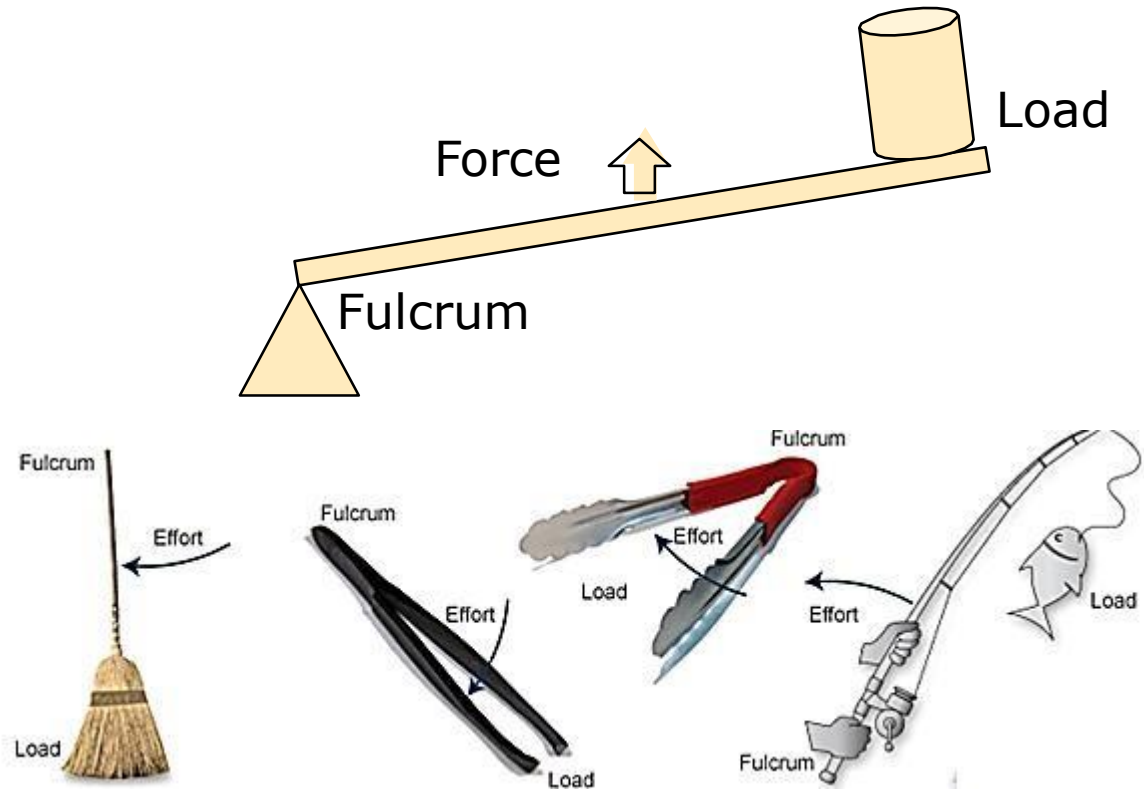
Second class lever: In the second-class lever, the load is in the middle. An example of a second-class lever is a bottle opener. The load is the cap of the bottle, the fulcrum is the end of the opener, and the effort is made at the other end of the opener. When the handle is pulled upward, the cap is pushed open. The nutcracker and wheelbarrow are also second-class levers.



THIRD-CLASS LEVER

Third-class lever

Third-class lever: The third-class lever has the effort in the centre. This lever has a disadvantage as the effort used to lift the load is always greater than the load. Examples of a third-class lever are tweezers and tongs.



In a pair of tongs, the load is the object to be picked up, the fulcrum (support) is the part that joins the two arms, and the effort is made in the middle. When we press the two arms together, they grip the object, and we can lift it.

Simple machines	What are they?	Work they do	Examples
Inclined Plane	A flat slanting surface	Moves things up or down	Stairs, slide
Lever	A stiff bar that rests on a support called a fulcrum	Lifts or moves load	Shovel, nut-cracker, see-saw, tweezers
Pulley	A grooved wheel with a rope or cable around it	Moves things up, down or across	Crane, curtain, flag post

VOCABULARY

Simple BN machines	What are they?	Work they do	Examples
Screw	An inclined plane	Holds things together or lifts things	Screw , jackscrew
Wedge	An object with at least one slanting side forming a sharp edge	Cuts or spreads open an object apart	Knife, chisel, axe
Wheel and axle	GA wheel with a rod (axle) fitted through its Centre, that move together	Lifts or moves loads	Tap, steering wheel